

RIDGE WRIGHT

SECURITY-FOCUSED SYSTEMS ADMINISTRATOR

Identity & Access Management | Endpoint Management & Defender | Automation Engineering | Incident Response

SUMMARY

Systems Administrator transitioning into a Cybersecurity Analyst role, with hands-on experience across Microsoft Defender XDR threat hunting, Intune-based endpoint management, and enterprise identity and access management engineering. This portfolio highlights recent initiatives spanning identity modernization, endpoint hardening, automation, and incident response.

IDENTITY & ACCESS MANAGEMENT

- Designed and enforced Conditional Access policies requiring non-phishable MFA (hardware passkeys, Windows Hello for Business, Microsoft Authenticator passkeys, macOS Platform SSO) and managed-device compliance for all account sign-ins.
- Deployed Windows Hello for Business via Cloud Kerberos Trust (CKT) for hybrid-joined devices, standing up the CKT DC object in Active Directory and rolling it out organization-wide — enabling a shift to Entra-joined-only device deployments while preserving on-prem print server access.
- Remediated 186 mixed hybrid and Entra-only identities ahead of the CKT rollout, authoring a PowerShell script to reconcile Azure account attributes into Active Directory and reconstruct the correct anchor key for each user.
- Enabled LDAP signing across Active Directory to close an unauthenticated bind vulnerability.
- Migrated printer and internal web application authentication from Basic SMTP auth to OAuth via vendor and internally-owned Azure App Registrations.

ENDPOINT MANAGEMENT & DEFENDER

- Enrolled over 50% of organizational endpoints into Intune and Microsoft Defender for Endpoint.
- Led the organization's migration to Microsoft E5 licensing, performing a risk and cost assessment to justify the investment and unlock the full Microsoft Defender suite.
- Built the organization's Intune Device Configuration profiles from the ground up. Created a primary profile assembled from 8 Open Intune Template baselines, plus two additional profiles for super-user (local admin) and government lockdown (Wi-Fi and broadcast service restrictions) use cases.
- Created App Protection Policies for Microsoft 365 mobile apps to support the organization's BYOD program, pairing Mobile Application Management (MAM) with a Conditional Access policy requiring protected-app usage.
- Implemented Microsoft Defender for Endpoint and Defender for Cloud Apps controls to block unsanctioned generative AI platforms across the enterprise.

AUTOMATION ENGINEERING

- Engineered a fully automated employee onboarding pipeline using an Active Directory-joined Azure VM configured as a Hybrid Runbook Worker, connected via site-to-site VPN between SonicWall and Azure.
- Authored 60+ PowerShell scripts across Azure Automation Accounts and 9 Power Automate flows, using webhook triggers to route script execution based on onboarding stage and user department.

INCIDENT RESPONSE & THREAT HUNTING

- Investigated and remediated a security incident involving HTML smuggling and process-hollowing malware, using advanced KQL queries in Microsoft Defender XDR to trace malicious file origins and network activity.
- Led end-to-end remediation of a token hijacking breach, conducting incident forensics and deploying hardened Conditional Access policies (managed-device requirement, non-phishable MFA) to close the exposure.
- Authored a formal incident report covering executive summary, risk analysis, technical breakdown, recommended security controls, and technical/executive conclusions.

HOMELAB

Hardware & Infrastructure

- Primary Hypervisor: Custom-built Proxmox VE server provisioned with an Intel Core i7-13700K processor, 64GB DDR4 RAM, and 12TB of all-flash storage (4x 3TB SSDs) for high-performance virtualization.
- Container Host: Custom-built Debian Docker server housed in a rackmount server chassis (repurposed Dell OptiPlex internals). Powered by an Intel Core i7-11700, 32GB DDR4 RAM, a 1TB NVMe/SSD for the OS/active containers, and a 3TB HDD for bulk storage.
- Edge Firewall: Custom-built OPNsense appliance featuring an AMD Ryzen 7 processor, 32GB DDR4 RAM, and a 1TB SSD for high-throughput routing, packet inspection, and local logging.
- Active Directory When needed in a Proxmox VM, connected to a personal Azure Cloud tenant.

Virtualization & Containerization

- Proxmox VE Environment: Provisions isolated virtual machines for core infrastructure and sandbox environments, including an NGINX reverse proxy, a TP-Link Omada network controller, a Windows testing environment, and offensive/privacy-focused security VMs (Kali Linux, Whonix-Gateway, and Whonix-Workstation).
- Docker & Portainer Ecosystem: Centralized container management hosting automation, IoT, and application services. Deployments include n8n for workflow automation, Home Assistant for smart home telemetry, and a dedicated Valheim server.

Networking & Security Operations

- Perimeter Defense & Routing: OPNsense acts as the primary edge device managing DHCP/DNS, utilizing CrowdSec for collaborative intrusion prevention and ClamAV for integrated antivirus scanning.
- DNS Filtering & Sinkholing: Network-wide ad blocking and malicious domain filtering managed via AdGuard Home.
- SIEM & Threat Monitoring: Deployed a full single-node Wazuh SIEM stack (Manager, Indexer, and Dashboard) via Docker to centralize log ingestion, vulnerability detection, and active threat monitoring across the environment.
- Secure Remote Access: Configured and maintain OpenVPN (OVPN) tunnels for secure, encrypted administrative access to the internal homelab network from remote locations.
- Run a dedicated DMZ for testing web hosting configurations and practice network isolation and segmentation.

ACRONYMS KEY

- BYOD (Bring Your Own Device) – A policy allowing employees to use their personal devices for work purposes.
- CKT (Cloud Kerberos Trust) – A Microsoft deployment model enabling Windows Hello for Business authentication for hybrid devices.
- DC (Domain Controller) – A server that processes security authentication requests within a Windows domain.
- DDR4 (Double Data Rate 4) – A generation of fast, high-bandwidth computer memory.
- DHCP (Dynamic Host Configuration Protocol) – A network management protocol used to automatically assign IP addresses to devices.
- DMZ (Demilitarized Zone) – An isolated subnetwork that exposes external-facing services to an untrusted network (like the internet) for added security.
- DNS (Domain Name System) – The system that translates human-readable web domain names into IP addresses.
- HDD (Hard Disk Drive) – Traditional magnetic storage hardware used for bulk data storage.
- HTML (HyperText Markup Language) – The standard markup language used to create and structure web pages.
- IoT (Internet of Things) – The network of interconnected physical devices and sensors that communicate and exchange data.
- KQL (Kusto Query Language) – A specialized query language used to search and analyze data in Microsoft security and analytics products.
- LDAP (Lightweight Directory Access Protocol) – An open protocol for accessing and maintaining distributed directory information over a network.
- MAM (Mobile Application Management) – Software and policies used to provision, secure, and control access to corporate mobile applications.
- MFA (Multi-Factor Authentication) – A security mechanism requiring two or more forms of verification to access an account.
- NVMe (Non-Volatile Memory Express) – A high-performance storage protocol designed to accelerate data transfer speeds for flash drives.
- OAuth (Open Authorization) – An open standard for access delegation and token-based authentication.
- OS (Operating System) – The core system software that manages a computer's hardware and software resources.
- OVPN (OpenVPN) – An open-source virtual private network protocol used to create secure, encrypted network connections.
- RAM (Random Access Memory) – The physical hardware inside a computer that temporarily stores active, working data.
- SIEM (Security Information and Event Management) – A centralized solution that provides real-time analysis of security alerts and log data across an environment.
- SMTP (Simple Mail Transfer Protocol) – The standard communication protocol used for sending and routing email.
- SSD (Solid State Drive) – A high-speed data storage device that uses flash-based memory instead of spinning disks.
- SSO (Single Sign-On) – An authentication method allowing users to log in once with a single set of credentials to access multiple independent systems.
- VE (Virtual Environment) – A networked operating environment that hosts virtualized infrastructure (as used in Proxmox VE).
- VM (Virtual Machine) – A software-based digital emulation of a physical computer system.

- VPN (Virtual Private Network) – A service that establishes a secure, encrypted connection across a public or shared network.
- XDR (Extended Detection and Response) – A cybersecurity platform that natively integrates multiple security products into a cohesive threat monitoring and response system